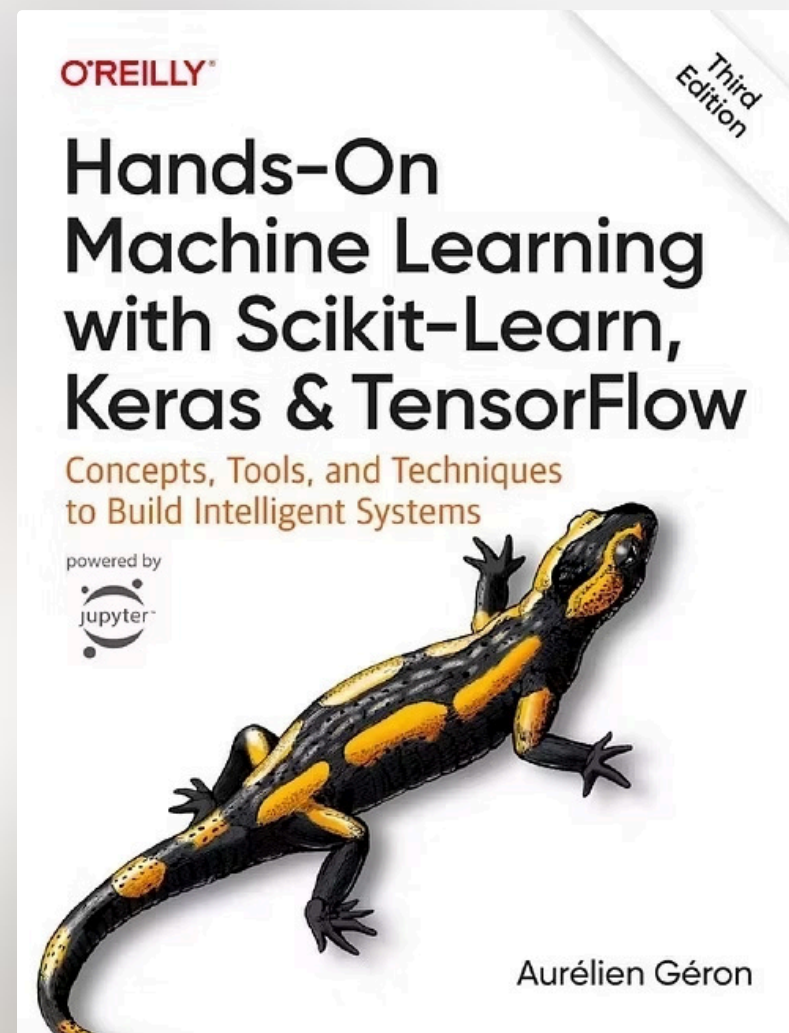


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

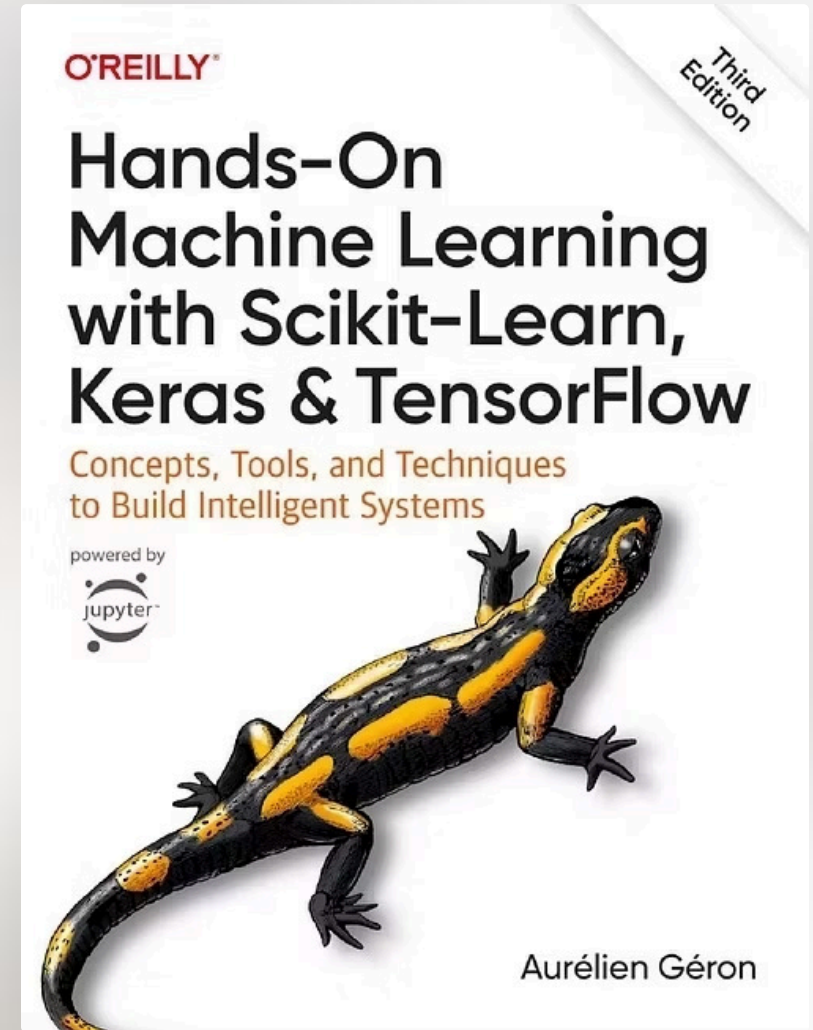
RP by Rodrigo Martins Pagliares

Last update: 07/01/2025



Machine Learning

Chapter 1 – The Machine Learning Landscape





Agenda for Chapter 1

1. What Is Machine Learning?
2. **Why Use Machine Learning?**
3. Examples of Applications
4. Types of Machine Learning Systems
5. Main Challenges of Machine Learning
6. Testing and Validating

Agenda for Chapter 1

1. What Is Machine Learning?

2. Why Use Machine Learning?

3. Examples of Applications

4. Types of Machine Learning Systems

- Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
- Classification criteria #2: Approach to Learning Method
 - Batch and Online (Incremental) Learning
- Classification criteria #3: Learning Method
 - Instance-Based and Model-Based Learning

5. Main Challenges of Machine Learning

- Insufficient Quantity of Training Data
- Non-representative Training Data
- Poor-Quality Data
- Irrelevant Features
- Overfitting the Training Data
- Underfitting the Training Data
- Stepping Back

6. Testing and Validating

- Hyperparameter Tuning and Model Selection
- Data Mismatch



Why use Machine Learning?



Consider how you would write a spam filter using traditional programming techniques...

The Traditional Approach Using Programming Techniques

Your program will likely become a **long list of complex rules**—pretty hard to maintain.

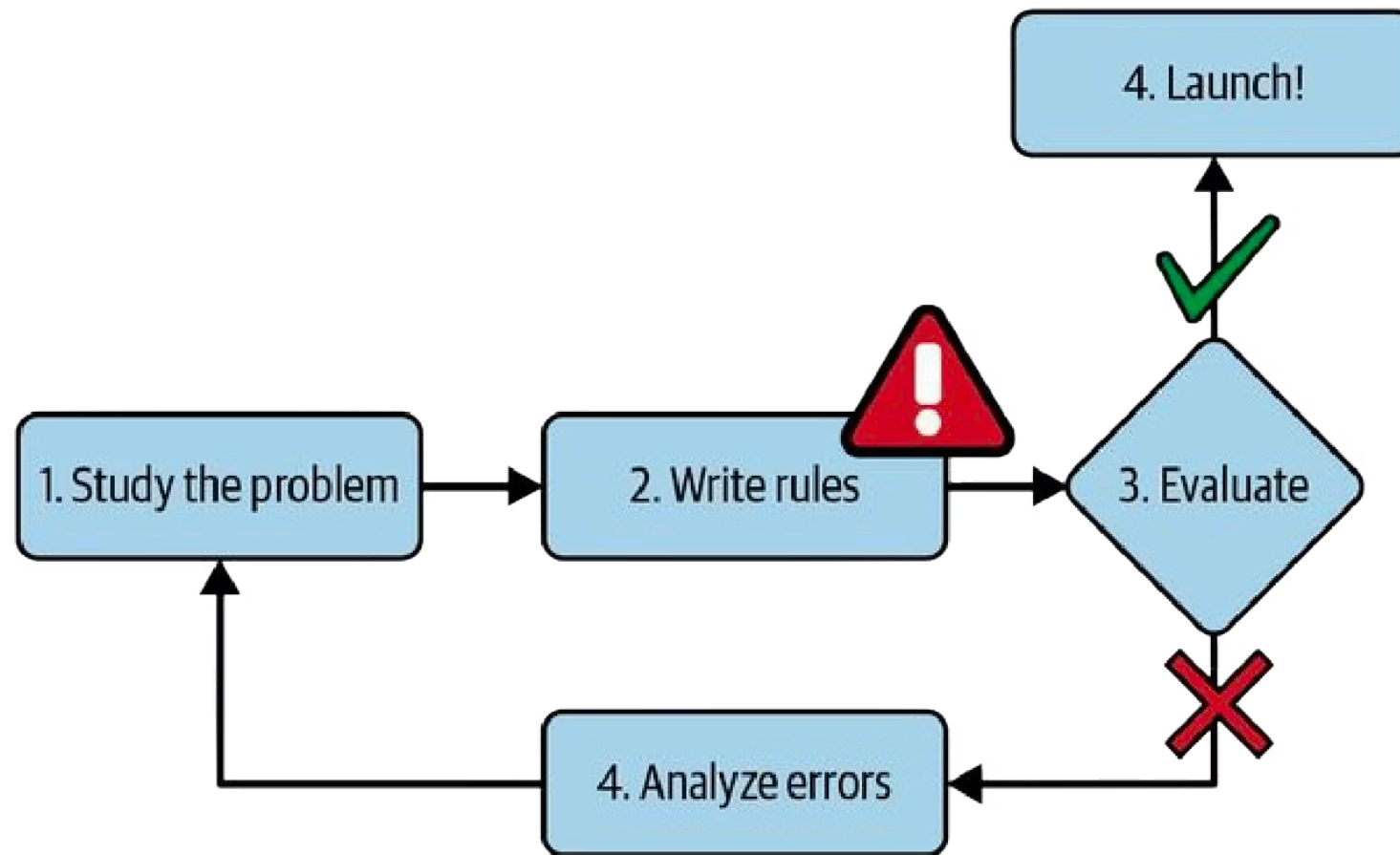


Figure 1-1. The traditional approach



Consider how you would write a spam filter using the ML techniques...

The ML Approach

- **ML enables automatic detection of spam indicators.**
 - Identifies frequent word patterns in spam vs. ham emails
 - Learns predictive phrases from labeled examples
 - **Reduces** manual rule creation and maintenance effort
 - Resulting program is shorter, easier to update, and more accurate

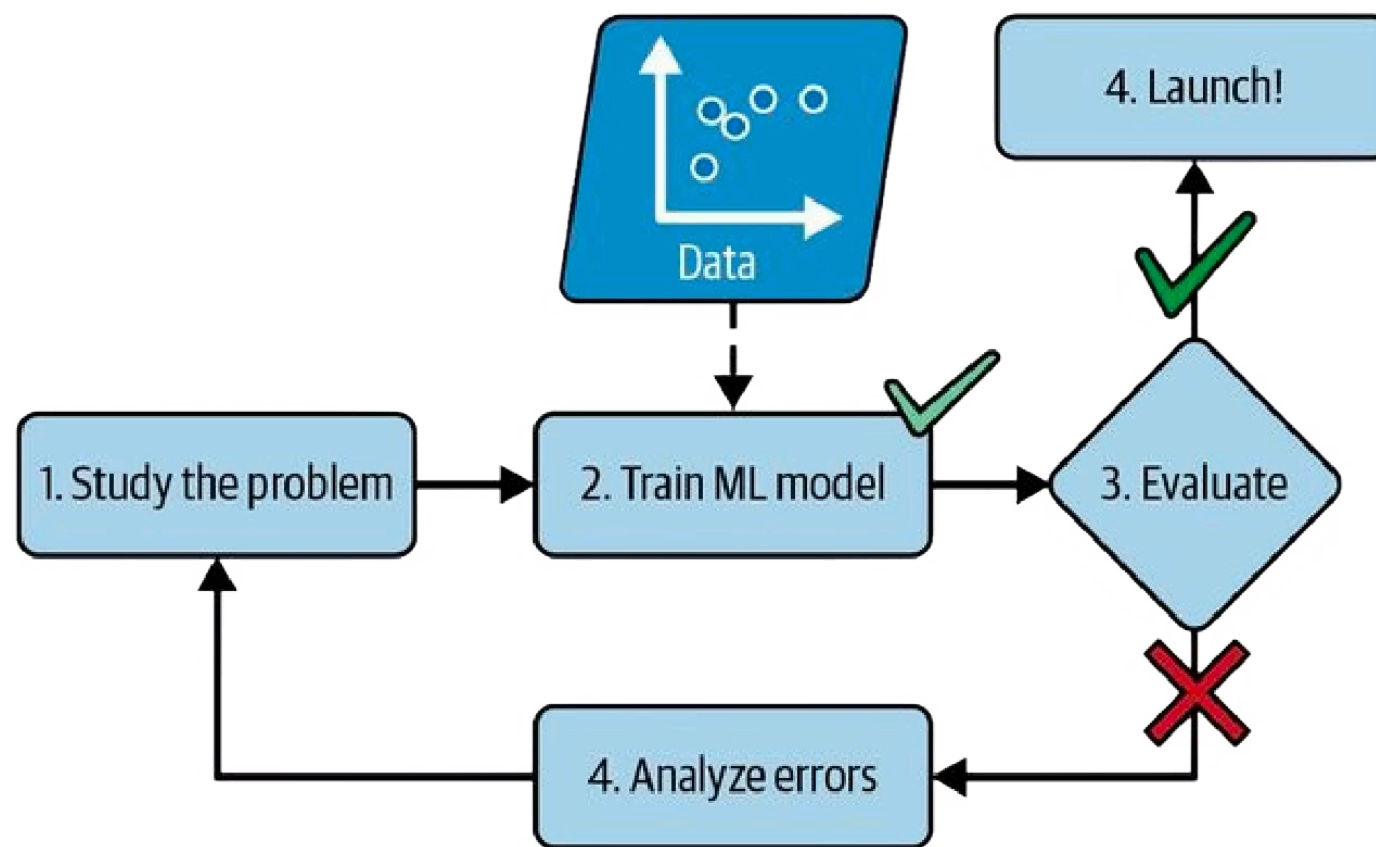


Figure 1-2. The machine learning approach

The ML Approach

- **ML solutions** can automatically adapt to changes
- What if **spammers** notice that all **their emails** containing **"4U"** are **blocked**?
 - They might start writing **"For U"** instead.
 - A **spam filter** using **traditional programming** techniques would **need to be updated** to **flag "For U"** emails.
- If **spammers** keep working around your **spam filter**, you will need to **keep writing new rules forever**.

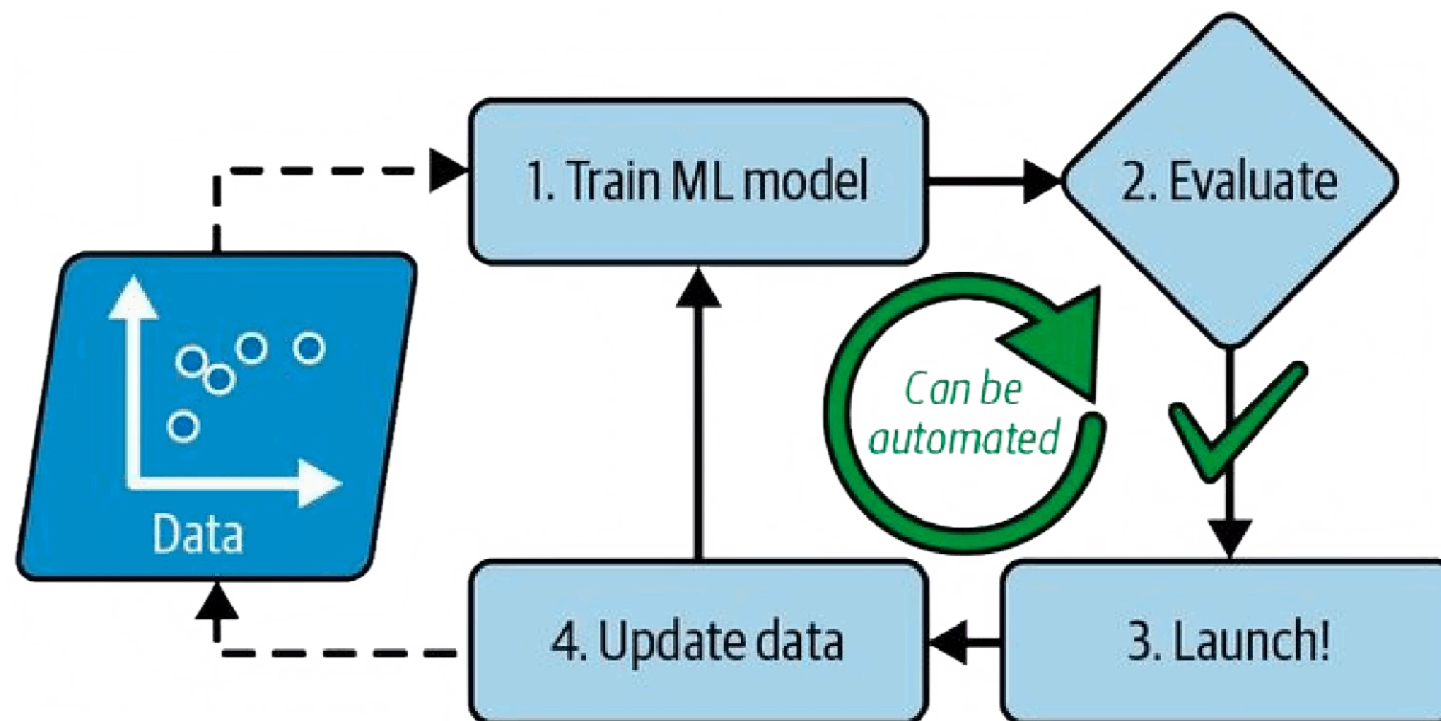


Figure 1-3. Automatically adapting to change

Note: A **spammer** is a **person, program, or entity** that **sends unsolicited, irrelevant, or inappropriate messages** to a **large number of recipients**, usually for **advertising, phishing, or spreading malware**.

The ML Approach

- **ML solutions can help humans learn.**
- **Trained filters can expose key terms strongly linked to spam**
 - **Reveals unexpected associations** and emerging spam trends
 - Offers deeper **insights** into **how spam messages evolve**
- **Uncovering such patterns from large datasets is known as data mining**
 - **ML is particularly effective** at this **exploratory task**

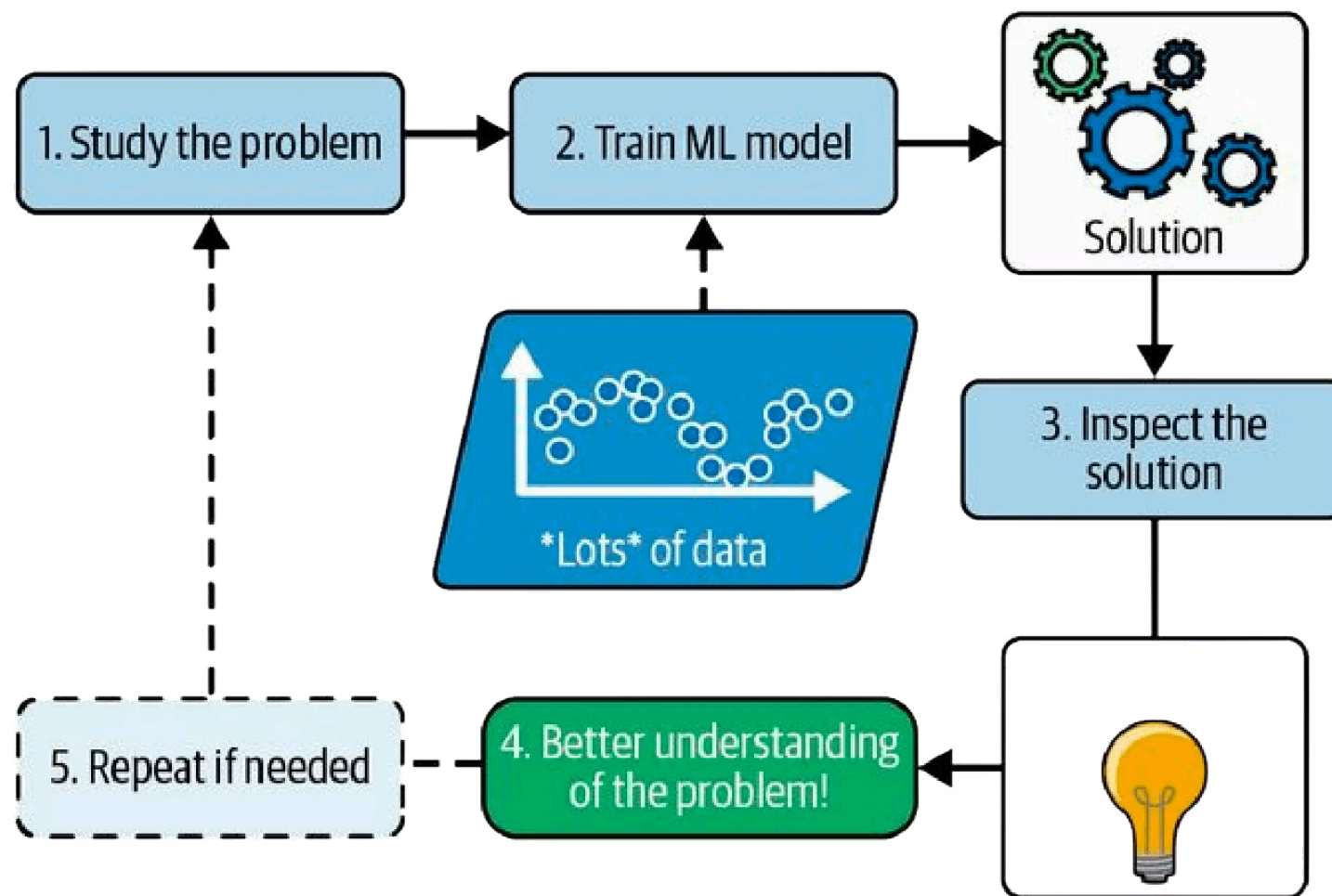


Figure 1-4. Machine learning can help humans learn

Summary: Why Use ML?

Traditional programming

Requires coding explicit rules for every situation (e.g., spam detection using hardcoded keywords).



Machine Learning

Automatically adapts to new patterns in data

- e.g., spam filters learn new terms like "For U").

Ideal for complex, dynamic problems with no known algorithms

- e.g., speech recognition.

Data mining: ML models help uncover hidden patterns in large datasets (e.g., customer behavior, fraud detection).



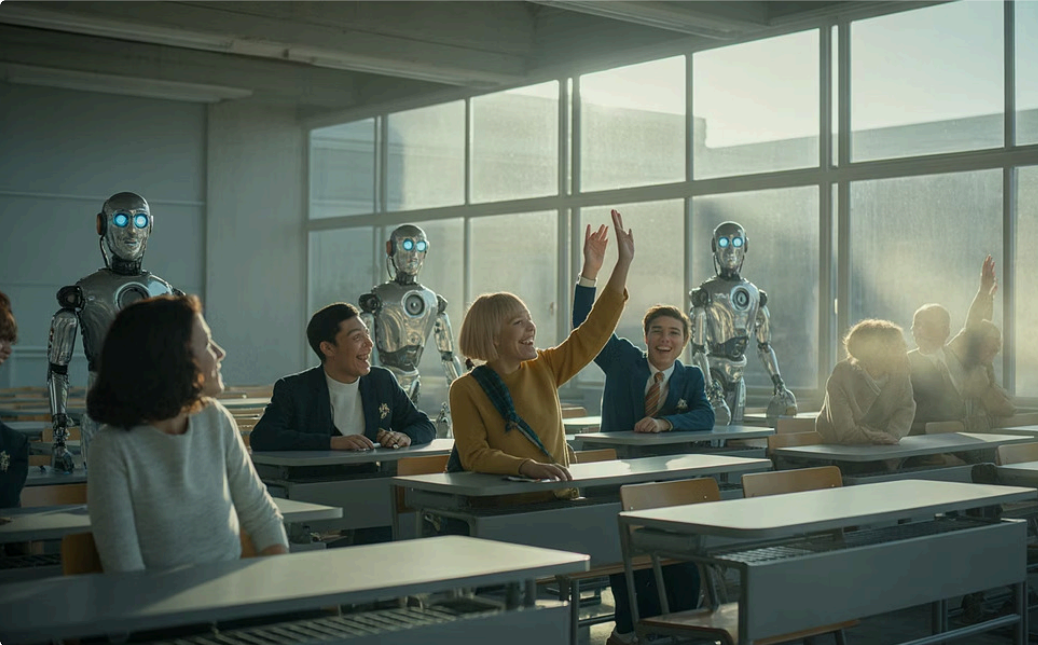
Summary: Why Use ML in a Spam Filter?

"Because it can automatically learn to recognize patterns in emails and accurately classify them as spam or not based on examples, without being explicitly programmed with hardcoded rules."

Next: Examples of Applications

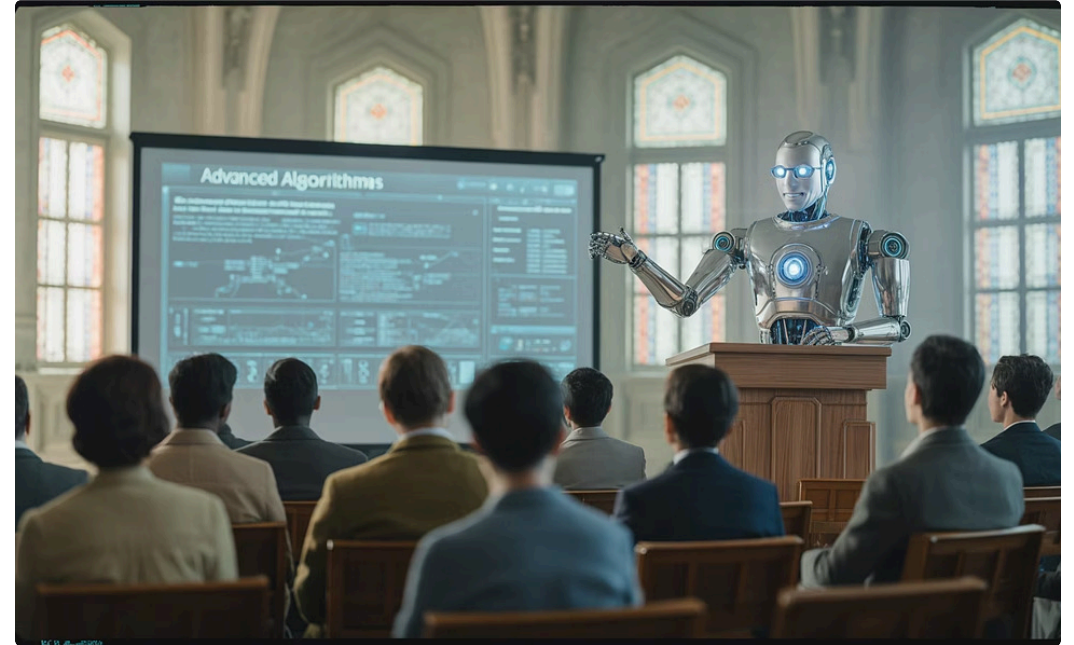


Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts